Measuring the Direction of Innovation

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Abstract

Most measures of the direction of innovation rely on patent data, which capture only a small fraction of firm-level innovation activity. We construct a comprehensive firm-level dataset of recently introduced products and processes by prompting a large language model with a uniform set of innovation-survey-style questions for approximately 600,000 firms across more than 40 countries, recovering nearly 6 million distinct introductions. Production-function regressions on the merged firm-financials sample show that the LLM-derived innovation indicators carry economically meaningful signal beyond labor, capital, and sector-by-country controls: a one-standard-deviation increase in the product (process) innovation score is associated with about a 5% (2.5%) higher TFP. Classifying each introduction as green, labor-augmenting, or labor-saving, we document four patterns. First, LLM-based measures detect product innovation in over 40% of firms and process innovation in nearly 30%, against patent incidence below 4%. Second, firms are predominantly engaged in labor-augmenting innovation—24% of firms on the product side and 15% on the process side—with labor-saving shares roughly half as large. Third, green innovation is similarly under-represented by patents (fewer than 1% of firms have a Y02 patent) but visible in 24% of firms via product-side LLM indicators and 15% via process-side indicators; cross-country greenness rankings nonetheless correlate strongly across the two measurement approaches. Fourth, there is substantial cross-country and cross-sector heterogeneity in greenness and in the augmenting–saving balance, with Korea and Japan among the greenest economies, China leading on green product introductions, and Software & IT exhibiting the smallest augmenting margin in process innovation.

Keywords: Innovation direction; Green innovation; Automation; Labor bias; LLMs; Patents
JEL codes: O30; O31; O33; O34; Q55; J24

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1 Introduction

Understanding the quality of economic growth, and not only its quantity, is an increasing concern for both economic policy and academic inquiry. The two key areas of concern are the "greenness" and the inclusiveness of growth. The dramatic growth since the industrial revolution has been associated, by and large, with a rapid increase in greenhouse gases and other potentially catastrophic impacts on the natural environment. The last decades have seen growth whose benefits were increasingly unequally distributed.

Long-run growth is driven by productivity and innovation, but its welfare consequences depend on the *direction* that innovation takes – in particular, whether innovation is on balance green or polluting, and whether it complements workers or substitutes for them. These two dimensions are the focus of this paper. Existing empirical work has measured this direction through several partial views of innovation activity. One branch uses patents to study environmental direction. Popp (2002) uses patent data to study energy-saving innovation, while Aghion et al. (2016) distinguish clean from dirty auto patents in a directed-technical-change framework. A broader clean-technology patent literature uses technology classes to measure climate-mitigation and renewable-energy innovation across countries and firms (Dechezleprêtre et al., 2011; Dechezleprêtre et al., 2013; Noailly and Smeets, 2015; Noailly, 2012). A second branch uses patents and tasks to study labor direction. The task-based framework of Acemoglu and Restrepo (2018) and Acemoglu and Restrepo (2019) distinguishes technologies that substitute for labor from technologies that create or reinstate labor tasks. Empirical work has operationalized this distinction by linking patent text to tasks, occupations, or job titles. Webb (2020) measures automation exposure from patent-task similarity. Mann and Püttmann (2023) train a classifier on manually coded patent text to identify automation patents. Dechezleprêtre et al. (2024) construct a firm-level measure of automation patents. Kogan et al. (2023) develop measures of labor-saving and labor-augmenting exposure using patents and job tasks, and Autor et al. (2024) link new occupational titles to patent-based measures of automation and augmentation.

Other work infers technology direction from non-patent sources, including job postings, occupational task data, and administrative data on technology adoption.[1] These sources are valuable, but they usually observe labor demand, occupational exposure, or a small set of

[1] Job-postings/vacancy-text approaches include Acemoglu et al. (2022b), who use Burning Glass postings to identify AI demand at the establishment level, and Babina et al. (2024), who use LinkedIn workforce data to link firms' AI hiring to product innovation. Task-level exposure measures based on O*NET task descriptions include Felten et al. (2021)'s AI Occupational Exposure index, the suitability-for-machine-learning score of Brynjolfsson et al. (2018), Eloundou et al. (2023)'s task-level LLM-exposure ratings, and Garg et al. (2026)'s country-conditioned automation-exposure measures; these are typically aggregated to occupations, industries, countries, or country-task cells rather than firms. Administrative-data work on technology adoption includes Acemoglu et al. (2022a) on the U.S. Census Annual Business Survey, Bessen et al. (2023) on Dutch firm-level automation expenditure, and Aghion et al. (2024) on French firm panels.

administratively recorded technologies rather than broad firm-level evidence on innovation. Job postings and occupation-based measures often do not describe the underlying innovation directly. Patent data have the opposite limitation: they describe technologies in detail, but exclude innovation outside the patent system.

A complementary concern is how much innovation activity these standard windows, namely patents and formal R&D, actually capture. In the World Bank Enterprise Surveys analysed by Cirera and Maloney (2017), around half of firms in most countries report introducing a new product or process, while fewer than a third undertake any formal R&D. The innovation visible in patents or R&D is therefore a small and selected slice of what firms are doing on the ground. Innovation surveys such as CIS and the World Bank Enterprise Surveys can in principle close this gap, but they are costly, infrequent, limited in geographic and sectoral coverage, and bounded in the number of innovations they record per firm. These literatures guide the classification we use below: green innovation follows the clean-technology literature, while labor-saving and labor-augmenting innovation follow the task-based automation and new-work literatures.

Thus, existing work is limited along various dimensions of comprehensiveness: the innovations captured, the types and number of firms captured, and the frequency and cost of information gathering. To overcome these limitations, we use LLMs to turn publicly available firm information into a comprehensive database of innovation activities for a sample of over 600,000 firms in more than 40 countries. The approach can easily be scaled to a larger sample of firms and countries. While a cross sectional exercise for now we can extend in time by repeatedly sampling going forward or by using historic snapshots of "the Internet" available in databases such as Common Crawl.[2]

For each firm in the sample, we repeatedly prompt an LLM with the same questions about recently introduced products and processes. We are effectively conducting an innovation survey (Mairesse and Mohnen, 2010) using an LLM. Unlike traditional innovation surveys, we are not limited to a small sample or to a few innovations per firm. In total, we record nearly 6 million newly adopted products or processes across approximately 600,000 firms.

To validate the approach, we merge the LLM-derived data with balance-sheet data and run production-function regressions. The LLM-based innovation indicators carry information about what makes firms more productive that goes beyond what is captured by standard production-function approaches with labor, capital, and sector and country controls. We find that a one-standard-deviation increase in an LLM-based product innovation indicator is associated with about 5% higher TFP, while a comparable process innovation indicator is associated with about 2.5% higher TFP.

We classify innovation into the following types: green, i.e., related to the mitigation or

[2] https://commoncrawl.org/

adaptation of climate change, labor–saving and labor improvement.[3] This leads to the following results.

First, LLM-based measures suggest dramatically more innovation than patent-based measures: we find evidence of product innovation in over 40% of firms and evidence of process innovation in nearly 30%, compared with evidence of patents in less than 4% of firms.[4]

Second, firms are predominantly engaged in labor-augmenting innovation. On the product side this amounts to 24% of firms and on the process side to about 15%. In each case, the labor-saving share is roughly half the augmenting share.

Third, like innovation overall, green innovation is under-represented by patents: fewer than 1% of firms have green (Y02) patents. Using LLM-based innovation indicators we find evidence of green product innovation in 24% of firms and green process innovation in 15%. That said, the share of green firms at the country, sector, and country-sector level correlates highly across LLM-based and patent-based green tagging.

Fourth, there is substantial variation in the augmenting bias and the greenness of innovation across countries and sectors. On greenness, some of the usual suspects top the rankings, such as Korea and Japan. China leads on green product innovation but ranks lower on green process innovation. Across sectors, Software & IT unsurprisingly comes last on augmenting process innovation; even so, (slightly) more firms there engage in labor-augmenting than in labor-saving innovation.

Overall, this paints an optimistic picture regarding the inclusivity of technical change going forward and is in contrast to widespread fears of job losses due to emerging AI technologies. If most innovations are focused augmenting rather than replacing labor then the fruits of economic growth arising from these innovations should be more evenly distributed. However, this message comes with some caveats. Like with patent data we might be making a mistake by treating all innovations and all firms as equal in economic value or their consequence on economic welfare and its distribution. Certain technologies - take for instance key platform or general purpose technologies - might be able to give a disproportionate opportunity to the people who control these technologies to extract value from the rest of the population. A related caveat is that our measures are necessarily conditioned on firms having an online footprint: countries and sectors where firms maintain less public web material will look less innovative under this lens than they may in fact be, and the cross-country comparisons should be read

[3] Green refers to innovation related to climate mitigation or adaptation. Labor–saving refers to innovation that automates tasks or reduces labor requirements, while labor–augmenting refers to innovation that complements workers or raises their productivity. These labels are assigned to each extracted innovation using a common LLM prompt, with details in Appendix A.

[4] The LLM-based shares are in the same neighborhood as the WBES benchmark of around 50% of firms reporting a product or process introduction (Cirera and Maloney, 2017), against R&D incidence below 30%; the order of magnitude separating LLM- and patent-based measures is therefore not an artifact of the LLM extraction.

accordingly. In future work we will have to address these issues.

The rest of the paper is organized as follows: Section 2 will discuss our basic data and measurement infrastructure. Section 3 discusses our key validation exercise that consists of including LLM derived innovation measures into a production function regression. Section 4 discusses how we tag innovations as either labor augmenting, labor saving or green and shows statistics on the prevalence of firms engaged in these various innoaviton types. We also examine how these innovation types vary between countries and sectors and how they relate to each other. Section 5 concludes.

2 Data and Measurement

The starting point is a firm sample drawn from Orbis. We first draw from a capped Orbis country sample and then retain firms for which the LLM extraction pipeline can be run on publicly available firm information. Within the sampled firms, those that meet the minimum financial-data requirements (positive sales and employment, valid country and sector codes, non-missing capital or a flag for missing capital) form what we call the *regression sample* of about 309,000 firms. We then use a broader *full snapshot* of about 605,000 firms for analyses that require only the LLM innovation measures, country and sector information, and patent data. The full snapshot is not a balanced country-stratified draw from Orbis: country counts reflect the original capped draw, active-status filtering, public-information availability, and financial-data coverage. We therefore report the regression sample and the full snapshot side by side throughout.

Two features of this sampling frame matter for interpretation. The full snapshot covers Orbis-registered firms rather than all firms in a country, so cross-country comparisons should be read as patterns across the formal, Orbis-visible firm population. Orbis coverage of small and informal firms is itself uneven across countries, and we expect this to register as conservative innovation incidence where these segments are under-represented. The LLM extraction could in principle recover innovation activity for firms outside Orbis, since it needs only a firm name and country to retrieve and read publicly available material. We restrict the sampling frame to Orbis here so that the resulting indicators can be merged with balance-sheet data for the production-function validation in Section 3.

For each firm we then prompt an LLM repeatedly with the same set of innovation-survey-style questions, asking it to list recently introduced products and processes it can find evidence of in publicly available information. Each individual introduction returned by the LLM is then classified along three dimensions: whether it is *green* (related to climate-change mitigation or adaptation), whether the labor effect is *augmenting* or *labor-saving* (or unclear), and whether the introduction is on the product or process side. We aggregate up from each individual

introduction to firm-level binary tags: a firm is tagged as "green-product" if it has at least one green product introduction, "augmenting-process" if it has at least one labor-augmenting process introduction, and so on. This gives us six firm-level direction tags (green, augmenting, labor-saving, each on the product and process side) plus the union "any green" across the two sides. Appendix A summarizes the extraction and classification protocol.[5]

To benchmark these LLM-derived measures against the patent record, we use Orbis-attributed patent counts with PATSTAT 2021 Spring as the underlying source and the firm–patent mapping coming from OrbisIP. "Clean" patents are those in CPC class Y02 (climate-mitigation technologies). Patent counts are stocks through approximately 2020 and are therefore older than the 2022–2025 window covered by the LLM extraction; we treat this as a feature rather than a bug, since the LLM measures recent activity and the patent stock measures the technological capability accumulated up to the point that activity was undertaken.[6]

When we aggregate firm-level direction tags to a country (or sector, or country$\times$sector cell), we report the fraction of firms in that geography carrying the tag, weighted equally across firms. For cross-country and cross-sector visualisations we restrict to geographies with at least 25 firms in the regression sample (or at least 100 for sector aggregations), and we use the same country/sector set across the regression-sample and full-snapshot views so that the two visualisations are directly comparable. Headcounts by country across the four overlapping subsets used in the analysis (full snapshot, with-text, regression sample, and the intersection) are reported in Appendix Table A1.

3 Production Function Regression

Before turning to the direction-of-innovation patterns that are the substantive focus of the paper, we want to establish that the LLM-extracted product and process content carries economically meaningful information about firms' productive performance. The concern is straightforward: an LLM prompted with a question about a firm's recent innovations may simply reproduce generic boilerplate about an industry or country, in which case the resulting text would be a noisier proxy for sector-by-country fixed effects rather than a genuine firm-level innovation

[5]The public material the LLM reads is multilingual: it consults firm websites and other public sources in the local language where available and English-language firm material where not, while the extraction and classification prompts themselves are in English. We do not find systematic differences in extracted innovation incidence between English-language and non-English-language jurisdictions of comparable Orbis coverage; a more careful audit of language-related extraction quality is left for follow-up work.

[6]We restore the green and total patent counts from a pre-winsorisation snapshot of the Orbis-attributed counts. Our headline analysis parquet winsorises patent counts at the regression-sample 99th percentile, which incidentally sets the green-count column to zero for every firm because fewer than 1% of firms have any green patent. The pre-winsor counts give a usable green-share measure for cross-country comparisons.

indicator. The natural test is to ask whether the text content of a firm's introductions predicts its sales beyond what is already explained by standard production-function controls (labor, capital, sector, country, and patent stock).

To set up this test we construct two firm-level innovation scores. For each of the product and process sides separately, we train a TF-IDF lasso on the text of the extracted introductions to predict log sales, after partialing out log employment, log capital, a capital-missing indicator, log(1+official patent count), and country×NACE2 fixed effects. To prevent the score from mechanically fitting the outcome it is then asked to explain, we use $K = 3$ cross-fitting: each firm's score is the held-out prediction from a lasso trained on the two folds that do not contain it. Firms with no extracted text in a given category have a score of exactly zero by construction, and an indicator for "has any extracted text" absorbs the corresponding level shift.

Two diagnostics anchor the score before we use it. Figure 1 plots residualized log sales against each cross-fitted innovation score: both panels show a clean positive relationship, but the relationship is meaningful precisely because the score is held-out from the firm whose sales it is being correlated with. Appendix Figure A1 renders the fold-averaged lasso coefficients as a word cloud. The terms that show up loaded most heavily are recognizable innovation vocabulary.

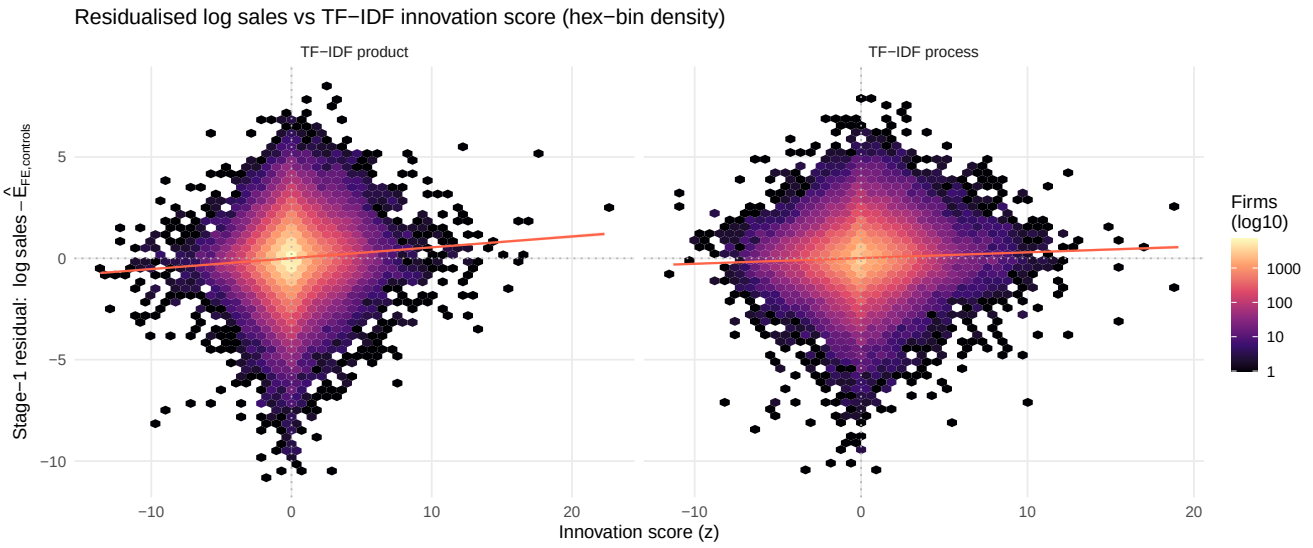


Figure 1: Stage-1 residualized log sales vs cross-fitted TF-IDF innovation score. Product (left) and process (right). Hex-bin density of all text-bearing firms; red line is an OLS fit with a 95% confidence band. Stage-1 residualization regresses log sales on log employment, log capital, the capital-missing flag, the official-patent indicator, and country×NACE2 fixed effects.

Table 1 reports the production-function regressions. Column (1) is the non-text baseline: log employment, log capital, the capital-missing indicator, log(1+official patent count), and country×NACE2 fixed effects. Columns (2) and (3) add the cross-fitted product and process

innovation scores separately, each with its has-text indicator and the corresponding count regressor (log(1+number of product introductions) in column (2); log(1+number of process introductions) in column (3)). Column (4) includes both scores and both count regressors together. The patent and size controls enter symmetrically in every column. Standard errors are two-way clustered at country and NACE2.

Three patterns stand out. First, both the product and the process innovation scores carry significant explanatory power for log sales conditional on patents, size, and country$\times$NACE2 fixed effects: the LLM-extracted text content is informative about firm productivity beyond what the standard production-function inputs capture. Second, the process score is roughly half as large as the product score in column (4) but remains highly significant. Third, the patent coefficient is positive and significant throughout, but its magnitude attenuates as the text scores enter — consistent with patents and LLM-extracted text capturing partly overlapping innovation signal. In quantitative terms, a one-standard-deviation increase in the product (process) innovation score is associated with about a 5% (2.5%) higher log sales, conditional on labor, capital, patents, and country-by-sector fixed effects. We treat this as evidence that the LLM extraction is doing real work and turn to its substantive content next.

Table 1: Production function regressions. Dependent variable: log revenue (log of firm sales). Sample: regression-sample firms with positive sales and employment.

	(1) Baseline	(2) + Prod score	(3) + Proc score	(4) + Both
Product innovation score (z)		0.056***		0.053***
		(0.008)		(0.008)
Process innovation score (z)			0.029***	0.024***
			(0.004)	(0.004)
Has any product text		0.023		0.038
		(0.032)		(0.039)
Has any process text			−0.004	−0.010
			(0.028)	(0.036)
log(1 + official patent count)	0.032	0.032	0.033	0.033
	(0.023)	(0.022)	(0.023)	(0.022)
log(1 + product introductions)		−0.015		−0.043
		(0.046)		(0.056)
log(1 + process introductions)			0.031	0.039
			(0.036)	(0.049)
Log employment	1.015***	1.014***	1.015***	1.014***
	(0.041)	(0.041)	(0.041)	(0.041)
Log capital	0.153***	0.152***	0.153***	0.152***
	(0.032)	(0.032)	(0.032)	(0.032)
Capital missing	1.578***	1.573***	1.576***	1.571***
	(0.348)	(0.346)	(0.347)	(0.346)
Num.Obs.	309 222	309 222	309 222	309 222
R2	0.861	0.861	0.861	0.861
R2 Adj.	0.860	0.860	0.860	0.860

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Notes: Standard errors two-way clustered at country and NACE2 in parentheses. The product and process innovation scores are K=3 cross-fitted TF-IDF lasso predictions of log-revenue residuals (after partialling out the controls and fixed effects), constructed in the companion paper. The scores are zero by construction for firms with no extracted text in the relevant category; the has-text indicators absorb the corresponding level shift between text-bearing and no-text firms.

4 Greenness and Net Labor Direction

We now turn to the substantive content of the LLM extraction. Each individual introduction has been classified along three direction tags — green, labor-augmenting, and labor-saving — and the question is whether those tags pick up genuine direction-of-innovation patterns or simply reflect classifier idiosyncrasies. A useful first sanity check is to ask which words actually populate each tag. Figure 2 reports a contrastive bigram analysis on the ∼5.9 million extracted introductions: for each tag, bigrams are scored by their log-odds-with-Dirichlet-prior z-score (Monroe et al., 2008), which measures how distinctively a phrase appears in that tag relative to the rest of the corpus. The top-ranked bigrams are then rendered as a word cloud with size and color proportional to the z-score.[7] The three tags read very differently. The green tag is dominated by phrases such as "cold chain", "waste reduction", "electronic invoicing", and "predictive maintenance" — a mix of resource-efficiency and digital-process themes that aligns naturally with what one would expect a green-classified introduction to be about. The labor-augmenting tag surfaces decision-support and copilot vocabulary; the labor-saving tag surfaces robotic process automation, self-service, and "unattended" execution patterns. The classifier appears to be picking up real semantic distinctions, not noise.

With the tag content sanity-checked, Figure 3 reports the headline composition of firm-level direction tags. The figure has three panels — product, process, and patents — and within each panel a faded bar reports the regression-sample share and a solid bar reports the full-snapshot share, so the reader can see how the picture changes as we move between the financials-conditional view and the broader coverage-aware view. The most striking feature of the figure is the gulf between the LLM-based and the patent-based view of innovation: roughly 40% of firms have at least one extracted product introduction and nearly 30% have at least one extracted process introduction, against patent incidence below 4%. Within the LLM-based tags, augmenting introductions are the most common direction tag on both sides, with labor-saving shares roughly half as large; green tags are smaller in absolute terms but still cover a non-trivial fraction of firms. As expected, the full-snapshot shares are mechanically lower than the regression-sample ones because the full denominator includes firms with no extracted text.

Does the direction of innovation varies systematically across countries. Figure 4 plots, for each country, the share of firms with at least one green introduction (x-axis) against the net-labor direction — the difference between the augmenting and the labor-saving firm-shares (y-axis) — separately for product introductions (left) and process introductions (right). Sample is the full 605k snapshot, and dot size is total firm count in the country.

[7] We make an anonymized version of the full database of all extracted product and process introductions available at www.prinzproject.io/innobase.

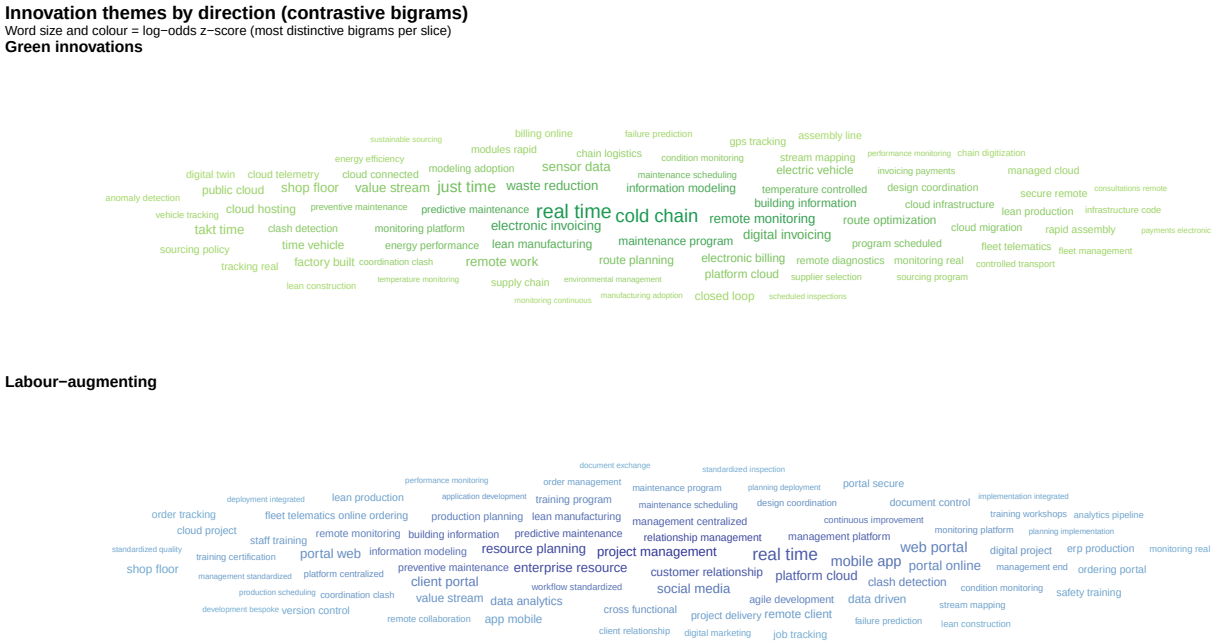


Figure 2: Contrastive bigrams characterizing the three direction-tag slices. For each slice (green, labor-augmenting, labor-saving) bigrams are scored by their log-odds-with-Dirichlet-prior z-score (Monroe et al., 2008), comparing bigram frequency in the slice to bigram frequency in the rest of the 5.9-million-event corpus. The top-80 bigrams by z-score are shown per panel; word size and color intensity are both proportional to the z-score.

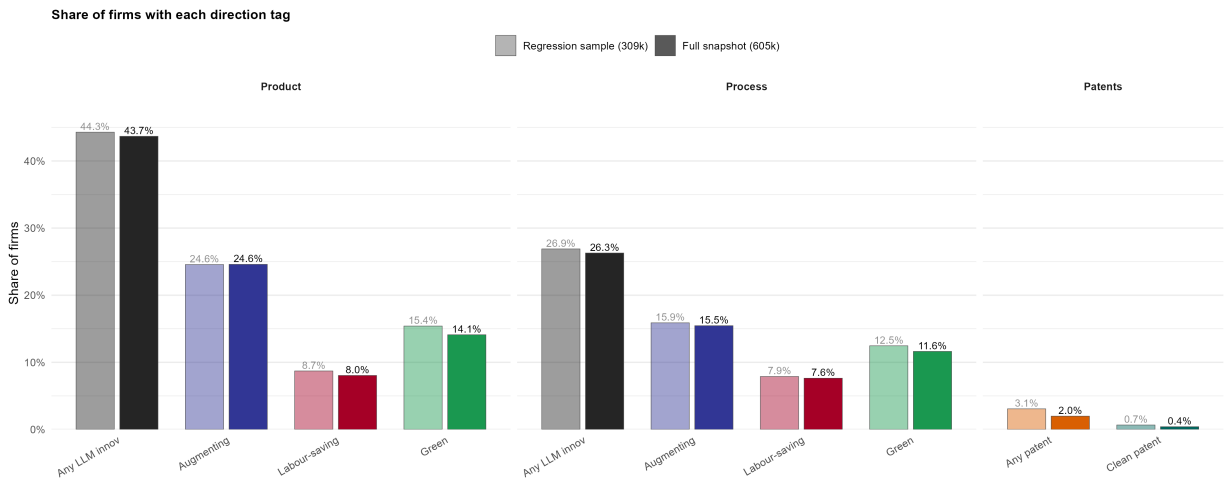


Figure 3: Share of firms carrying each direction tag. The figure has three panels: "Product" (left) reports the share of firms with any extracted product introduction in 2022–2025 ("Any LLM innov") alongside the share with a labor-augmenting, labor-saving, or green product introduction; "Process" (middle) reports the corresponding shares on the process side; "Patents" (right, narrower) reports the firm-level share with any official Orbis-attributed patent ("Any patent") and the share with at least one "clean" patent (CPC class Y02, climate-mitigation technologies). Within each tag, the faded bar is the regression sample (309k firms with required financials) and the solid bar is the full 605k snapshot. The full-snapshot bars are mechanically lower than the regression-sample bars because the full denominator includes firms with no extracted text and – in the patent panel – a substantial share of small firms outside the regression sample for which patent filings are rare.

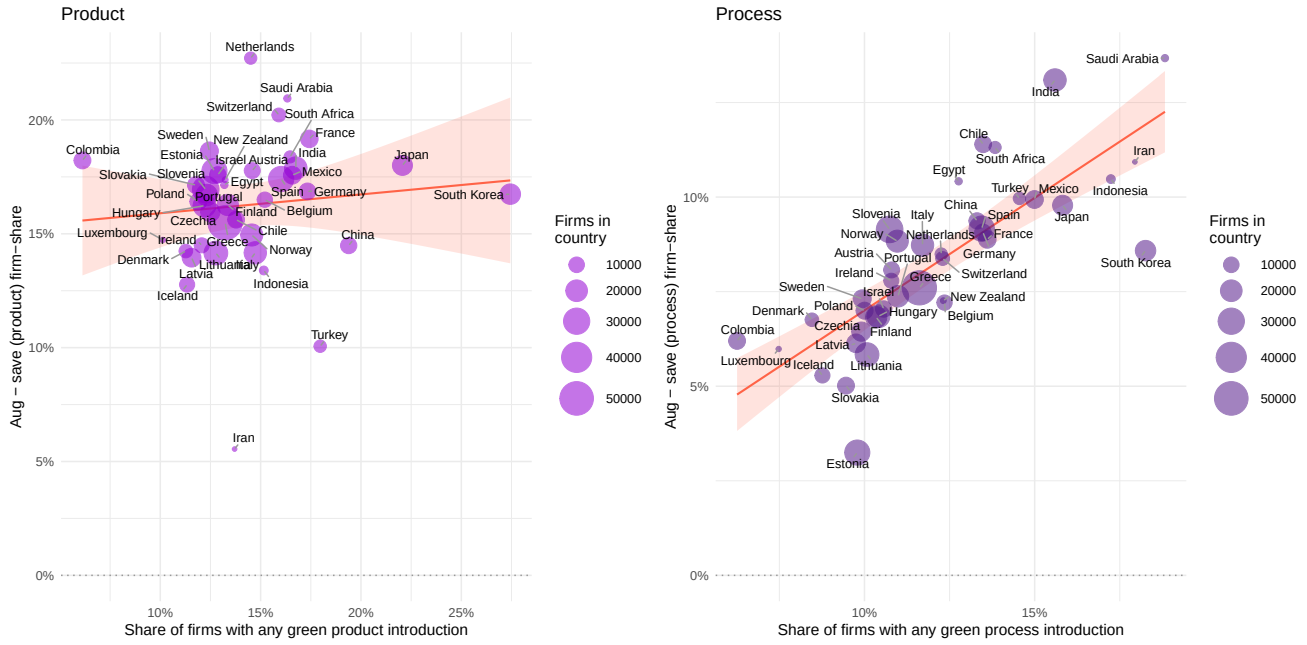


Figure 4: Direction of innovation: green firm-share vs net-labor direction (augmenting − saving), product (left) and process (right). Full 605k snapshot. Country labels at full names; red line is an OLS fit with shaded 95% confidence band.

Notes: Each dot is a country. The x-axis is the share of firms with at least one extracted green product (left) or process (right) introduction in 2022–2025. The y-axis is the difference between the augmenting-firm share and the labor-saving-firm share, where "augmenting" and "labor-saving" are LLM-classifier tags applied to each extracted introduction and aggregated to firm level (the firm carries a tag if it has at least one such introduction). Sample restricted to countries with at least 25 firms in the regression sample. Firms with no extracted text are included in the denominator and contribute zeros to the numerator.

Three patterns emerge. First, cross-country dispersion in green firm-shares is large: the spread on the x-axis runs over an order of magnitude. Second, the cross-country relationship between greenness and net-labor direction is positive in both panels — countries with greener product or process introductions also tilt their innovation toward labor-augmenting activities on net. Third, the relationship is tighter on the process side than on the product side. The fact that greenness and labor-augmentation co-move at the country level is, to us, a non-obvious finding; it suggests that the same kinds of firms (or country-level innovation environments) that lean toward climate-relevant innovation also lean toward complementing rather than displacing labor. For comparability, Figure 5 reports the same scatter on the 309k regression-sample firms; the cross-country ordering is preserved.

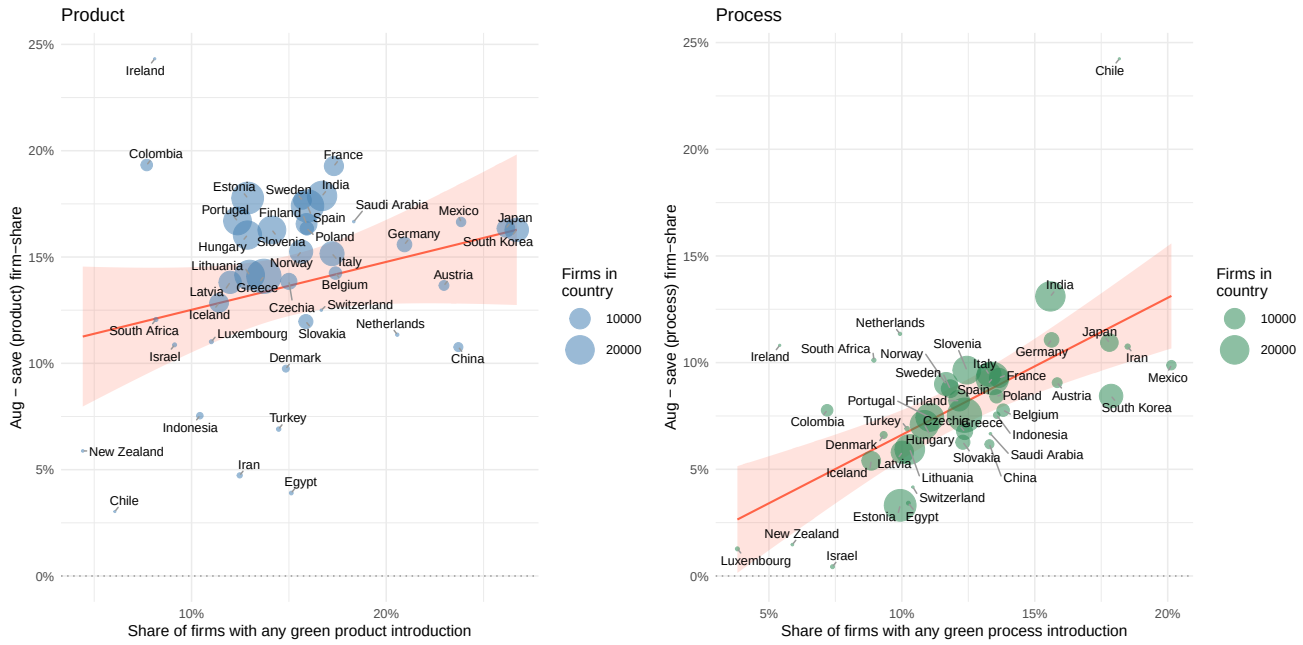


Figure 5: Direction of innovation, regression sample (309k firms with financials). Same construction as Figure 4.

Notes: Same as Figure 4 but restricted to the 309k regression-sample firms (firms with required financial fields). Country shares are mechanically larger because the denominator excludes firms with no extracted text.

The scatter plot is informative about the cross-country relationship but masks who actually sits at each end of the distribution. Figures 6 and 7 report top-10 / bottom-10 horizontal bar rankings of countries and NACE2 sectors on the net-augmenting firm-share and on the green firm-share respectively. Each figure has four subpanels: countries on the top row, NACE2 sectors on the bottom row; product introductions on the left, process introductions on the right. All rankings are computed on the full 605k snapshot. Korea and Japan rank at or near the top on green innovation, with China leading on green product innovation but ranking lower on green process innovation. On the labor-direction side, Software & IT predictably comes

last among sectors on net-augmenting process innovation — perhaps unsurprising given how much of recent process innovation in software is precisely about automating workflows — but, importantly, even there the firm-share engaged in labor-augmenting innovation slightly exceeds the firm-share engaged in labor-saving innovation. The country and sector sets are the same ones used in the body-text scatter plots: countries with at least 25 firms in the regression sample, and NACE2 sectors with at least 100 firms in the regression sample.

How much of the cross-country variation in Figures 6 and 7 reflects sectoral composition rather than within-sector country differences? Sector fixed effects explain more cross-firm variation than country fixed effects for every direction tag, with the gap largest on green-product (3.5 versus 0.8 percent) and smallest on augmenting-process (0.6 versus 0.5 percent), matching the wider dispersion of the sector panels (Appendix Table A2). The country rankings themselves are nonetheless stable under sector adjustment.[8] The largest adjustments hit industrial economies whose green and labor-saving tags partly reflect their sectoral mix, with Korea, Japan, Iran, Saudi Arabia, and Turkey appearing repeatedly as sector-inflated.

[8]Spearman rank correlations between raw and sector-adjusted country shares run from 0.89 on green-product to 0.995 on augmenting-process; see Appendix Figure A13.

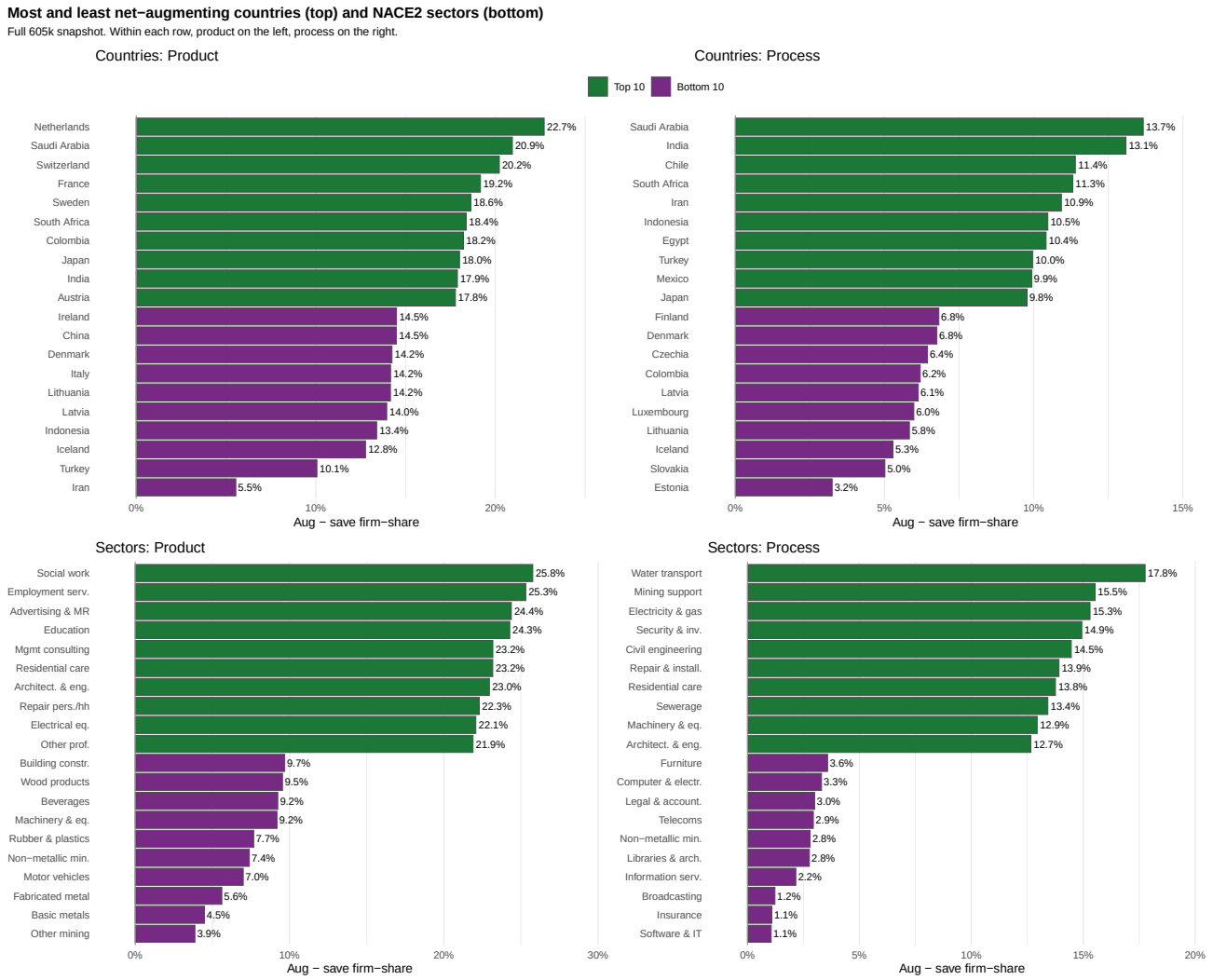


Figure 6: Top 10 and bottom 10 countries (top row) and NACE2 sectors (bottom row) by the augmenting–saving firm-share difference. Within each row, product (left) and process (right) panels rank independently. Top 10 in green, bottom 10 in purple. Computed on the full 605k snapshot.

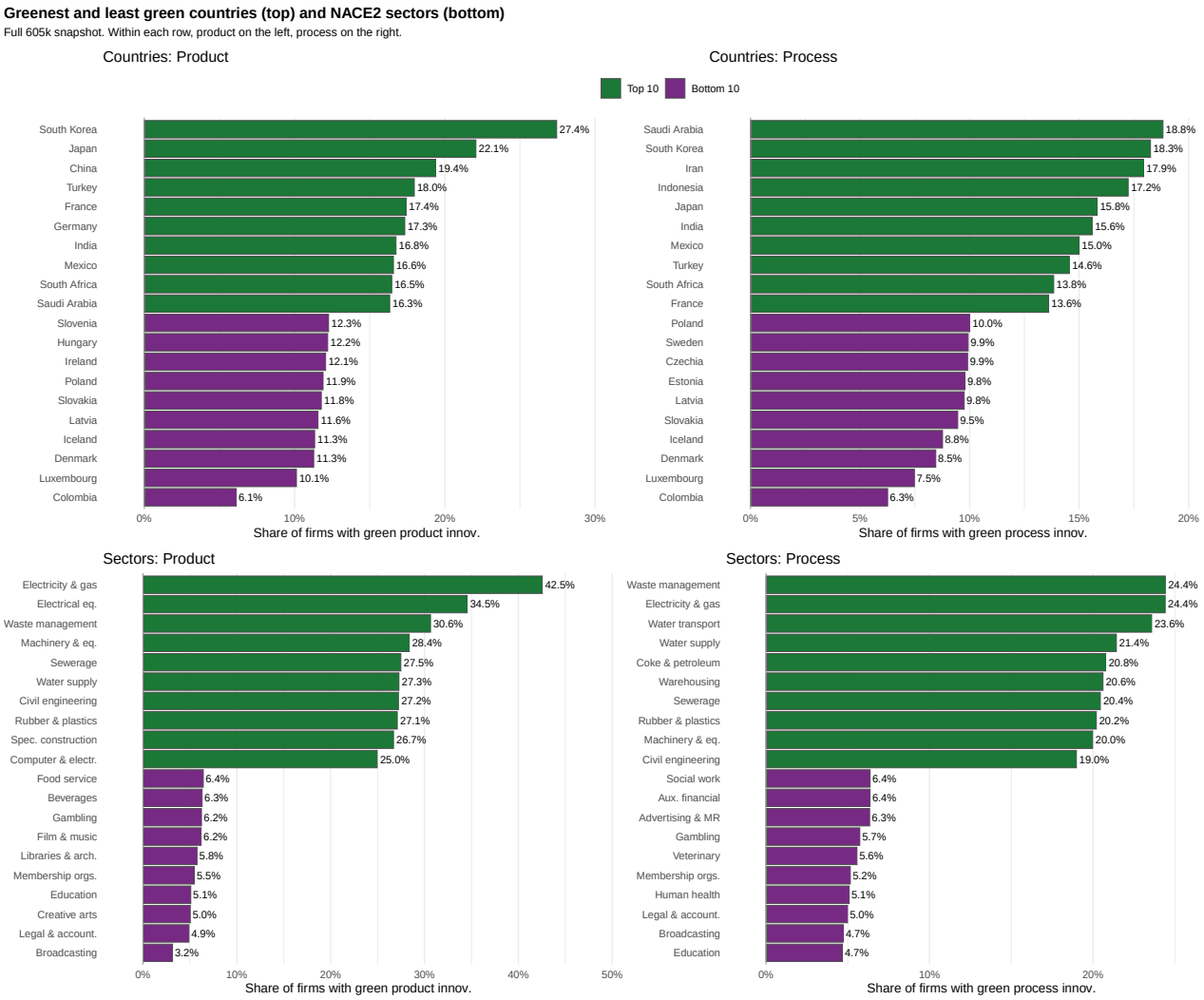


Figure 7: Top 10 and bottom 10 countries (top row) and NACE2 sectors (bottom row) by share of firms with any green product (left) or process (right) innovation. Within each row, product on the left, process on the right. Top 10 in green, bottom 10 in purple. Computed on the full 605k snapshot.

4.1 Augmenting vs Labor-Saving

The net-labor measure used so far compresses two underlying shares into a single difference, which is convenient for visualising directional patterns but can hide what is happening to each component on its own. Figure 8 unpacks the difference by plotting the augmenting firm-share against the labor-saving firm-share directly, with a 45-degree reference line indicating equality between the two. Two patterns are visible. First, the two shares co-vary strongly across countries on the product side and even more strongly on the process side: countries that innovate a lot tend to do so on both sides of the labor-direction divide, not exclusively in one. Second, however, the OLS fit lies almost everywhere below the 45-degree line, and the gap is widest at high innovation intensities. Countries with the highest augmenting-share have a substantially smaller labor-saving-share. This is the cross-country counterpart of the headline result in Figure 3: labor-augmenting innovation dominates labor-saving innovation, and the dominance is more pronounced precisely where overall innovation intensity is highest.

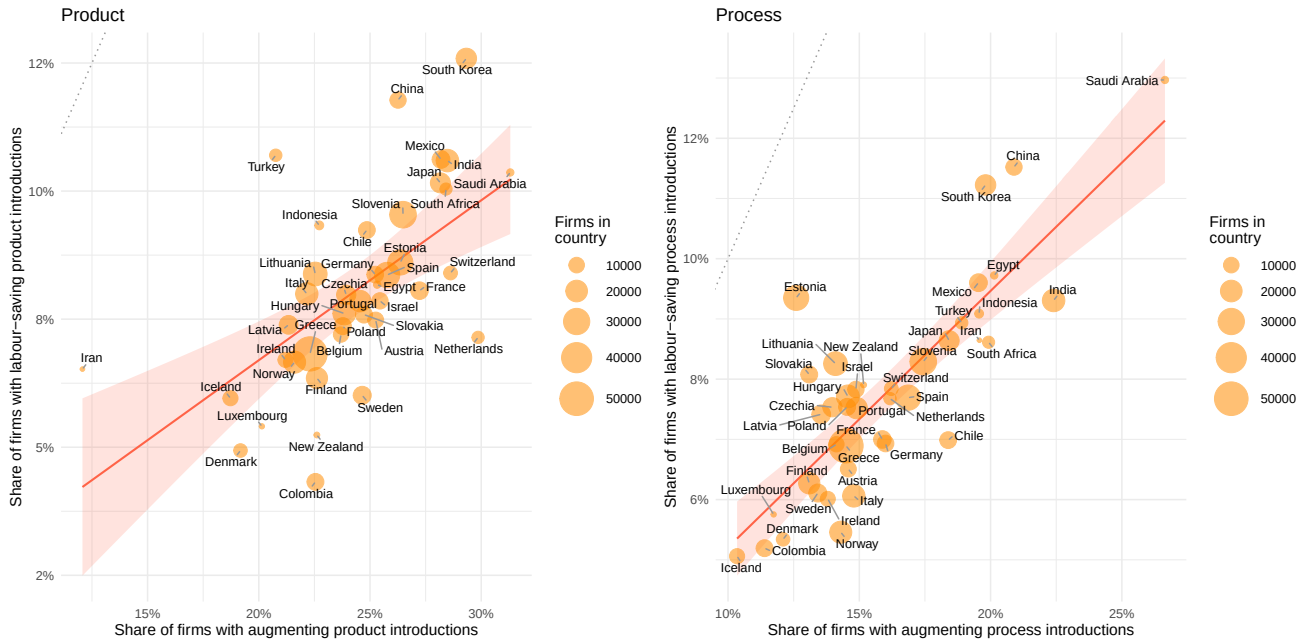


Figure 8: Augmenting vs labor-saving firm-shares by country, product (left) and process (right). Full 605k snapshot. Dotted line is 45-degree equality; red line is an OLS fit with 95% confidence band.

Notes: Each dot is a country. Augmenting and labor-saving firm-shares come from the LLM classifier's labor-effect label applied to each extracted introduction, aggregated to firm-level tags. Sample restricted to countries with at least 25 firms in the regression sample. Firms with no extracted text are included in the denominator and contribute zero to both numerators.

4.2 Greenness in the LLM Extraction vs Greenness in Patents

A central motivation of this paper is that patents capture only a small fraction of innovation activity. We can sharpen that point by comparing the LLM-based and patent-based view of greenness directly. Figure 9 plots, for each country, the share of firms with at least one green product or process introduction (x-axis) against the country's green patent share, defined as CPC Y02 patents over total patents (y-axis). The two panels report the same scatter on the regression sample (left) and the full snapshot (right); the country set is identical across the two panels.

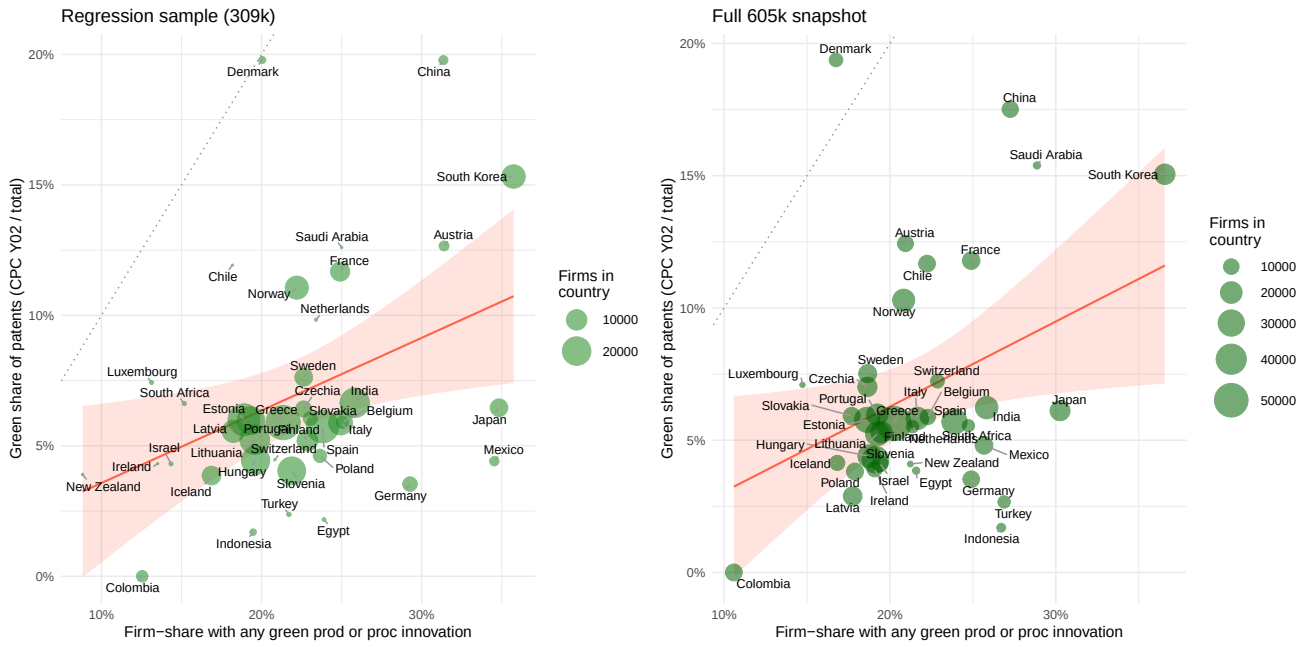


Figure 9: Green firm-share (any product or process) vs green patent share by country. Regression sample (left) and full 605k snapshot (right). Patent counts are the un-winsorised stocks from Orbis (firm-patent mapping via OrbisIP, PATSTAT 2021 Spring). Dotted line is 45-degree equality; red line is an OLS fit with 95% confidence band.

Notes: Each dot is a country. The x-axis is the share of firms in the country with at least one extracted green product or process introduction in 2022–2025. The y-axis is $\sum_f$ `off_patent_clean_count`$_f$ / $\sum_f$ `off_patent_count`$_f$, summed over firms in the country, with patent counts restored from the pre-winsorisation snapshot (the headline analysis parquet zeros out `off_patent_clean_count` via aggressive count winsorisation). Restricted to countries with at least 25 patents and at least 25 firms in the regression sample.

The cross-country relationship between LLM-extracted greenness and patent-attributed greenness is positive, and it tightens at the sector and country-by-sector level (Appendix Figures A7 and A12). The two measures are not orthogonal, and the patent ranking recovers the relative cross-country ordering. The OLS fit nonetheless lies well below the 45-degree line in both views: the LLM extraction identifies a substantially larger fraction of firms as engaged in green innovation than their patenting record does. This gap is the paper's "missing innovation" message in a direction-of-innovation context. Patents capture the concentration of frontier-style

green R&D; they understate how broadly firms engage in green activity, which is what matters for measuring the diffusion of climate-relevant innovation.

4.3 Cross-Sample Robustness

We have shown several patterns by reporting the regression sample and the full snapshot side by side, but a fair worry is that those two views are simply alternative numerical accounts of the same underlying object and that the cross-country direction story is being driven by whichever sample happens to be plotted. To address this we plot the two views directly against each other. Figures 10 and 11 put the regression-sample firm-share on the x-axis against the full-snapshot firm-share on the y-axis, separately for product and process and separately for greenness and the aug-save measure. If the cross-country direction story is robust, dots should line up on a near-monotone curve through the origin; if not, we should see scrambled rankings.

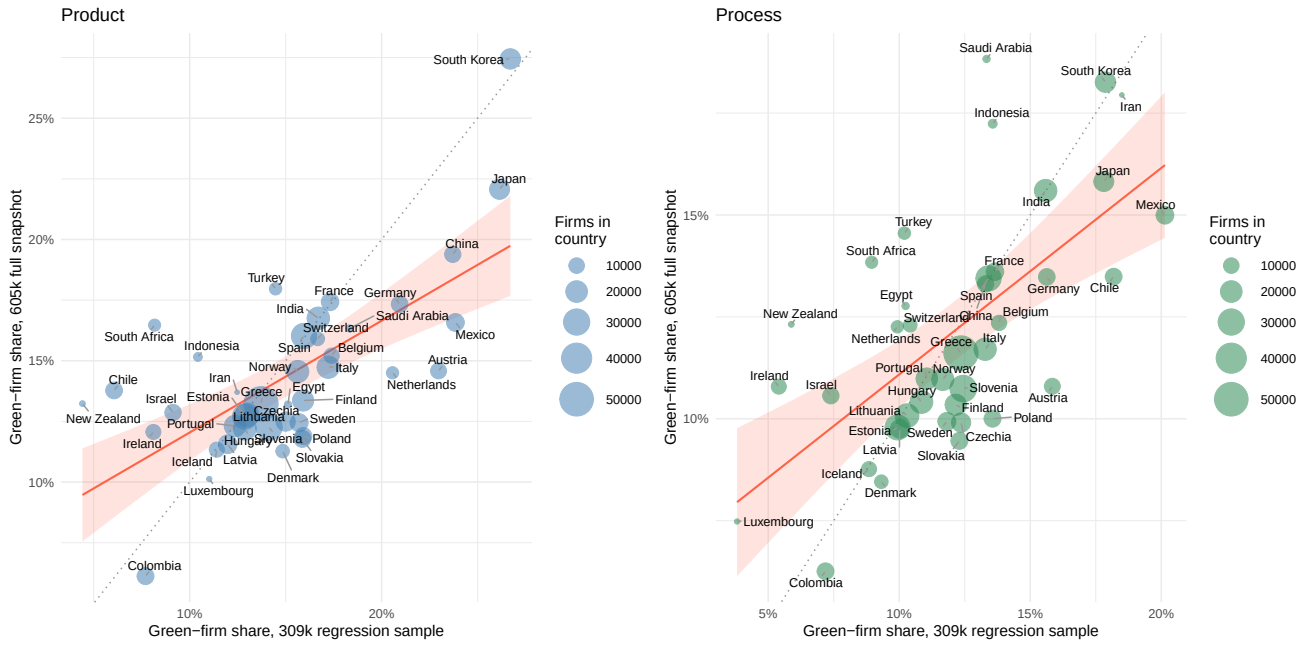


Figure 10: Cross-sample comparison: green firm-share in 309k regression sample (x-axis) vs the same measure in the 605k full snapshot (y-axis). Product (left) and process (right). Dotted line is 45-degree equality; red line is OLS fit.

Notes: Each dot is a country present in both samples. Below-the-line points reflect dilution: the full-snapshot denominator includes firms with no extracted text. The cross-country ranking is well preserved; the OLS slope reflects the average rate at which the regression-sample share shrinks when expanding to the full snapshot.

The two figures answer the worry differently but reassuringly. For the green firm-share, every country sits below the 45-degree line — because the full-snapshot denominator includes firms with no extracted text, the absolute level of the share is mechanically lower in the full sample. But the cross-country ordering is well preserved, and the OLS slope is well-estimated and below



Figure 11: Cross-sample comparison: aug-save firm-share in 309k regression sample (x-axis) vs the same measure in the 605k full snapshot (y-axis). Product (left) and process (right). Dotted line is 45-degree equality; red line is OLS fit.

Notes: Same construction as Figure 10 but for the aug-save measure (augmenting firm-share minus labor-saving firm-share). Because aug-save is a difference, the dilution effect is much weaker than for the green firm-share, so dots cluster more tightly around the 45-degree line.

one in a uniform proportional way. For the aug-save measure, which is a difference rather than a level, the dilution effect is much weaker (numerator and denominator scale together), and points cluster tightly around the 45-degree line. The bottom line: what changes between the two views is the level of country shares, not the relative cross-country ordering.

5 Conclusion

We set out to measure the direction of innovation at firm level for a much broader sample of firms and countries than patent or survey data alone permit, by treating an LLM as an instrument that can read the same publicly available information about each firm and produce a uniform innovation-survey-style record. The two motivating questions were whether innovation is, on balance, moving in a direction that helps offset pressures on climate and on inequality. The answer is a conditional yes on both counts: green innovation is far more widespread than patents suggest, and firms are predominantly engaged in labor-augmenting rather than labor-saving activity. Three substantive findings underpin this answer.

First, LLM-based measures detect dramatically more innovation than patent-based ones — product innovation in over 40% of firms, process innovation in nearly 30%, against patent

incidence below 4% — and the same gap re-appears when we look specifically at green innovation, with green tags assigned to a quarter of firms on the product side against fewer than 1% with a Y02 patent. Second, firms are predominantly engaged in labor-augmenting rather than labor-saving innovation, and this is true on both the product and process side and across countries and sectors with limited exceptions. Third, there is substantial cross-country and cross-sector heterogeneity that lines up with prior expectations along some dimensions (Korea and Japan among the greenest economies; Software & IT lowest on net-augmenting process innovation) and is informative along others (China leading on green product innovation but not on green process innovation; greenness and labor-augmentation co-moving systematically across countries).

If most innovation is augmenting rather than replacing labor, the fruits of innovation-driven growth should be more evenly distributed than recent commentary about emerging AI technologies has tended to assume. We view this as the most policy-relevant takeaway of the paper, and it is a takeaway that the patent literature does not have the granularity to make.

A few caveats are in order. We treat all firms and all innovations as equal in this analysis, but key platform or general-purpose technologies might give a disproportionate opportunity to whoever controls them to extract value from the rest of the economy — in which case a count-based view of innovation direction will mis-represent welfare consequences. The data are also currently a single cross-section: we plan to extend in time by repeatedly sampling going forward and by exploiting historic snapshots of public web content from sources such as CommonCrawl, which would allow direction patterns to be tracked over time and would open the door to identification strategies that exploit changes in firm-level innovation in response to policy. We leave these extensions for future work.

References

Acemoglu, Daron, Gary W. Anderson, David N. Beede, Catherine Buffington, Eric E. Childress, Emin Dinlersoz, Lucia S. Foster, Nathan Goldschlag, John C. Haltiwanger, Zachary Kroff, Pascual Restrepo, and Nikolas Zolas (2022a). *Automation and the Workforce: A Firm-Level View from the 2019 Annual Business Survey*. NBER Working Paper 28290. National Bureau of Economic Research. URL: https://www.nber.org/papers/w28290.

Acemoglu, Daron, David Autor, Jonathon Hazell, and Pascual Restrepo (2022b). "Artificial Intelligence and Jobs: Evidence from Online Vacancies". In: *Journal of Labor Economics* 40.S1, S293–S340. DOI: 10.1086/718327.

Acemoglu, Daron and Pascual Restrepo (June 1, 2018). "The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment". In: *American*

Economic Review 108.6, pp. 1488–1542. ISSN: 0002-8282. DOI: 10.1257/aer.20160696. URL: https://pubs.aeaweb.org/doi/10.1257/aer.20160696 (visited on 01/18/2026).

Acemoglu, Daron and Pascual Restrepo (May 1, 2019). "Automation and New Tasks: How Technology Displaces and Reinstates Labor". In: *Journal of Economic Perspectives* 33.2, pp. 3–30. ISSN: 0895-3309. DOI: 10.1257/jep.33.2.3. URL: https://pubs.aeaweb.org/doi/10.1257/jep.33.2.3 (visited on 01/18/2026).

Aghion, Philippe, Céline Antonin, Simon Bunel, and Xavier Jaravel (2024). "The Effects of Automation on Labor Demand". In: *Journal of the European Economic Association*. Forthcoming.

Aghion, Philippe, Antoine Dechezleprêtre, David Hémous, Ralf Martin, and John Van Reenen (Feb. 2016). "Carbon Taxes, Path Dependency, and Directed Technical Change: Evidence from the Auto Industry". In: *The Journal of Political Economy* 124.1, pp. 1–51. DOI: 10.1086/684581.

Autor, David, Anna Salomons, and Bryan Seegmiller (2024). *New Frontiers: The Origins and Content of New Work, 1940–2018*. Working Paper. NBER / Quarterly Journal of Economics. URL: https://www.nber.org/papers/w30389.

Babina, Tania, Anastassia Fedyk, Alex He, and James Hodson (2024). "Artificial Intelligence, Firm Growth, and Product Innovation". In: *Journal of Financial Economics* 151, p. 103745. DOI: 10.1016/j.jfineco.2023.103745.

Bessen, James, Maarten Goos, Anna Salomons, and Wiljan van den Berge (2023). "What Happens to Workers at Firms That Automate?" In: *Review of Economics and Statistics*. Forthcoming. DOI: 10.1162/rest_a_01284.

Brynjolfsson, Erik, Tom Mitchell, and Daniel Rock (2018). "What Can Machines Learn, and What Does It Mean for Occupations and the Economy?" In: *AEA Papers and Proceedings* 108, pp. 43–47. DOI: 10.1257/pandp.20181019.

Cirera, Xavier and William F Maloney (2017). *The Innovation Paradox: Developing-country Capabilities and the Unrealized Promise of Technological Catch-Up*. World Bank Publications.

Dechezleprêtre, Antoine, Matthieu Glachant, Ivan Haščič, Nick Johnstone, and Yann Ménière (2011). "Invention and Transfer of Climate Change–Mitigation Technologies: A Global Analysis". In: *Review of Environmental Economics and Policy*.

Dechezleprêtre, Antoine, Matthieu Glachant, and Yann Ménière (2013). "What Drives the International Transfer of Climate Change Mitigation Technologies? Empirical Evidence from Patent Data". In: *Environmental & Resource Economics* 54.2, pp. 161–178.

Dechezleprêtre, Antoine, David Hémous, Morten Olsen, and Carlo Zanella (2024). "Automating Labor: Evidence from Firm-Level Patent Data". In: *Journal of Political Economy*. Forthcoming.

Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock (2023). "GPTs Are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models". arXiv:2303.10130. URL: https://arxiv.org/abs/2303.10130.

Felten, Edward W., Manav Raj, and Robert Seamans (2021). "Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses". In: *Strategic Management Journal* 42.12, pp. 2195–2217. DOI: 10.1002/smj.3286.

Garg, Prashant, Tommaso Crosta, and Jasmin Baier (2026). "Global Automation Atlas". Working paper. URL: https://automationatlas.org/.

Kogan, Leonid, Dimitris Papanikolaou, Lawrence D. W. Schmidt, and Bryan Seegmiller (2023). *Technology and Labor Displacement: Evidence from Linking Patents with Worker-Level Data*. NBER Working Paper 31846. National Bureau of Economic Research. URL: https://www.nber.org/papers/w31846.

Mairesse, Jacques and Pierre Mohnen (2010). "Using Innovation Surveys for Econometric Analysis". In: *Handbook of the Economics of Innovation, Volume 2*. Vol. 2. Handbook of the Economics of Innovation. Elsevier, pp. 1129–1155. DOI: 10.1016/s0169-7218(10)02010-1.

Mann, Katja and Lukas Püttmann (2023). "Benign Effects of Automation: New Evidence from Patent Texts". In: *Review of Economics and Statistics* 105.3, pp. 562–579. DOI: 10.1162/rest_a_01083.

Monroe, Burt L., Michael P. Colaresi, and Kevin M. Quinn (2008). "Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict". In: *Political Analysis* 16.4, pp. 372–403. DOI: 10.1093/pan/mpn018.

Noailly, Joëlle (2012). "Improving the Energy Efficiency of Buildings: The Impact of Environmental Policy on Technological Innovation". In: *Energy Economics* 34.3, pp. 795–806.

Noailly, Joëlle and Roger Smeets (2015). "Directing Technical Change from Fossil-Fuel to Renewable Energy Innovation: An Application Using Firm-Level Patent Data". In: *Journal of Environmental Economics and Management* 72, pp. 15–37.

Popp, David (2002). "Induced Innovation and Energy Prices". In: *American Economic Review* 92.1, pp. 160–180.

Webb, Michael (2020). "The Impact of Artificial Intelligence on the Labor Market". SSRN working paper. DOI: 10.2139/ssrn.3482150. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3482150.

A LLM extraction and direction classification protocol

This appendix summarizes how we turn public firm information into the product, process, green, labor-augmenting, and labor-saving measures used in the paper. The same procedure is applied to every firm.

For each firm, we prompt the model with the firm's name and country and ask it to identify products and processes the firm has introduced since 2015. The model draws on publicly

available information about the firm in its training corpus.

The extraction prompt asks the model to identify concrete product and process introductions supported by the firm information bundle. The core instruction is:

> Read the information about the firm. Identify products and processes that the firm has recently introduced, launched, adopted, or substantially upgraded. Return only concrete introductions supported by the supplied information. Do not infer innovations from the firm's industry alone. Separate products from processes. For each item, provide a short description, and the introduction year if available.

The model returns a structured record with separate product and process entries. The main fields are:

Field	Description
`name`	Official product or service name (proper noun).
`general_description`	Concise, info-dense description of the introduction; brand-agnostic.
`release_year`	Introduction or release year (`YYYY`, 2015 onward; `NA` if unknown).
Firm-level fields:	
`sector`	Short sector label (2–5 words), e.g. "consumer electronics".
`employees_last_known`	Bucket: `1-5`, `6-20`, `21-100`, `101+`, or `NA`.
`hq_city`	Primary city where the firm is headquartered.

We classify an item as a product introduction when it is a new or substantially upgraded good or service offered by the firm. We classify an item as a process introduction when it is a new or substantially upgraded production, logistics, delivery, digital, or internal operating process used by the firm. Malformed records and duplicate records are removed in post-processing.

Each extracted introduction is then classified along the direction dimensions used in the paper. The core classification instruction is:

> Classify the introduction along the following dimensions. First, is it green, meaning related to climate-change mitigation or adaptation, energy efficiency, emissions reduction, renewable energy, recycling, or related environmental objectives? Second, what is its likely labor direction: labor-saving, labor-augmenting, or unclear? Use labor-saving when the introduction substitutes for labor, automates tasks, reduces labor requirements, or replaces worker activities. Use labor-augmenting when the introduction complements workers, raises worker productivity, improves decision-making, creates new tasks, or expands the scope or quality of work performed by labor. Use unclear when the description does not provide enough evidence.

The classification output is:

Field	Allowed values / description
`green`	Yes, no, or unclear.
`labor_direction`	Labor-augmenting, labor-saving, or unclear.
`classification_reason`	Short explanation for the assigned labels.

The main analysis aggregates these introduction-level records to firm-level indicators. A firm is coded as having green product innovation if at least one extracted product introduction is classified as green. A firm is coded as having labor-augmenting process innovation if at least one extracted process introduction is classified as labor-augmenting. We construct the same indicators for green, labor-augmenting, and labor-saving introductions separately for products and processes. Country, sector, and country-by-sector measures are firm-weighted shares of firms carrying each tag.

Firms with no extracted introduction are treated as zeros in the firm-share measures. These zeros mean that the pipeline found no public evidence of the relevant introduction under this protocol. They should not be read as definitive evidence that the firm undertook no innovation of any kind.

B TF-IDF score diagnostics

The cross-fitted TF-IDF innovation score used in Section 3 is built from a lasso of extracted-text TF-IDF features on residualized log sales. Figure A1 reports the fold-averaged lasso coefficients as a word cloud. The terms that load most heavily are recognizable innovation vocabulary, supporting the validation claim that the score is picking up information beyond country-by-sector boilerplate.

C Country sample composition

Table A1 reports firm headcounts per country across the four overlapping subsets we use throughout the paper: the full snapshot, the firms with any extracted text, the regression sample (firms with required financial fields), and the intersection. Sample restricted to countries with at least 25 firms in the regression sample, matching the country set used for the body-text scatter plots.

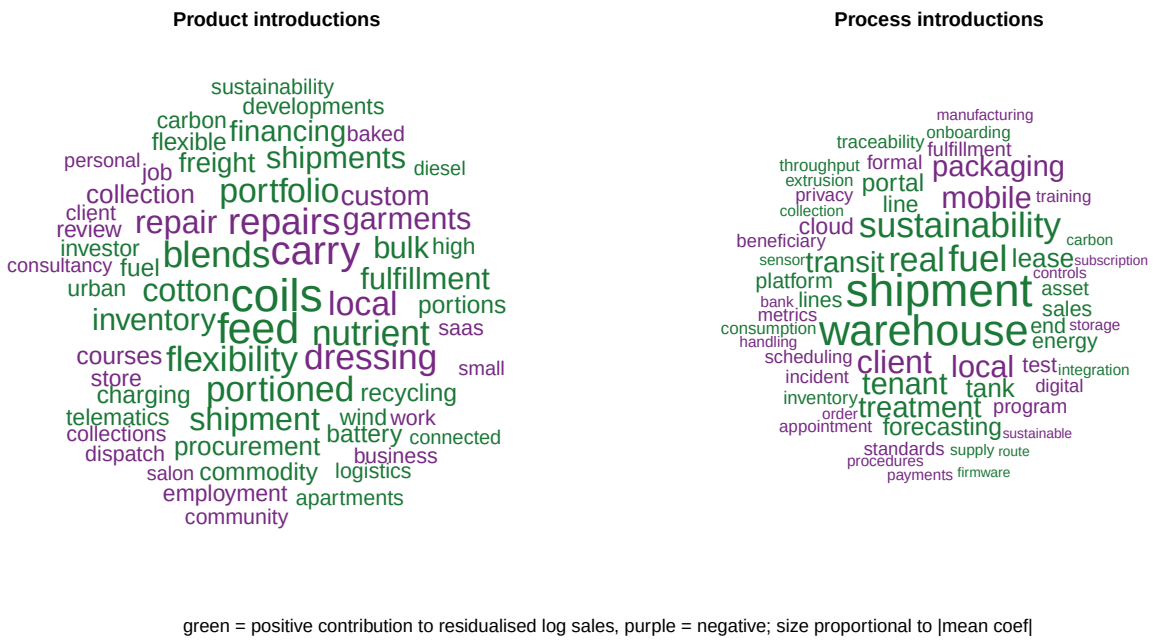


Figure A1: Word cloud of fold-averaged TF-IDF lasso coefficients, product (left) and process (right). Word size is proportional to coefficient magnitude; color encodes sign (green positive, purple negative).

Notes: Coefficients are averaged across the $K = 3$ cross-fit folds; the top 60 terms by absolute coefficient are shown per panel. The lasso is fit at the firm level on the extracted product or process text after partialing out log employment, log capital, the capital-missing indicator, $\log(1 + \text{official patent count})$, and country$\times$NACE2 fixed effects. Negative-loading terms enter with a negative coefficient on residualized log sales: after partialing out size and country-by-sector fixed effects, firms whose extracted text is heavier on those terms have lower sales than their controls would predict. Construction details follow the cross-fit pipeline in Section 3.

Table A1: Country headcount across the four overlapping subsets used in the analysis. *Total* is every firm in the analysis snapshot `analysis_w1_99.parquet`; *With text* restricts to firms with at least one extracted product or process introduction in 2022–2025; *In regression sample* restricts to firms with required financial fields (positive sales and employment, valid country/sector); the last column is the intersection. Sample restricted to countries with at least 25 firms in the regression sample.

ISO	Country	Total	With text	In regression sample	Text ∩ regression
GR	Greece	51877	24564	29143	14006
SI	Slovenia	30747	15557	18885	10088
ES	Spain	27623	14464	26022	13597
EE	Estonia	27327	13373	24947	12258
LT	Lithuania	24196	11986	21643	10825
HU	Hungary	22311	10356	19647	9329
IN	India	21651	13139	21267	12904
IT	Italy	21365	10432	13887	7053
NO	Norway	20793	9369	12827	5869
PT	Portugal	19574	9667	18866	9327
FI	Finland	19200	8458	10316	4863
KR	South Korea	17729	11386	13301	8447
JP	Japan	16958	9463	7267	4152
CZ	Czechia	15583	7546	5954	2936
LV	Latvia	14724	6631	11919	5344
SE	Sweden	13570	6183	7370	3564
MX	Mexico	13185	7830	1862	1227
FR	France	12860	6834	8693	4636
PL	Poland	11977	5623	4026	2006
SK	Slovakia	11885	5717	4530	2269
CO	Colombia	11842	4504	2949	1231
CL	Chile	11678	6150	33	12
DE	Germany	11469	5633	4862	2521
IL	Israel	11035	5516	230	84
CN	China	10971	7151	1700	1145
AT	Austria	10465	5211	2064	1099
BE	Belgium	9626	4606	3461	1660
IS	Iceland	9367	3580	8262	3152
IE	Ireland	9232	4069	37	13
CH	Switzerland	7701	4199	48	15
DK	Denmark	7415	3024	923	373
TR	Turkey	6122	3772	304	148
NL	Netherlands	6005	3163	141	61
ZA	South Africa	5817	3328	257	79
ID	Indonesia	2883	1736	796	342
EG	Egypt	1872	1086	205	131
SA	Saudi Arabia	1843	1203	60	28

Table A1: Country headcount across the four overlapping subsets used in the analysis. *Total* is every firm in the analysis snapshot `analysis_w1_99.parquet`; *With text* restricts to firms with at least one extracted product or process introduction in 2022–2025; *In regression sample* restricts to firms with required financial fields (positive sales and employment, valid country/sector); the last column is the intersection. Sample restricted to countries with at least 25 firms in the regression sample. *(continued)*

ISO	Country	Total	With text	In regression sample	Text ∩ regression
NZ	New Zealand	1088	532	68	13
LU	Luxembourg	869	381	236	93
IR	Iran	613	368	465	276

D Conditional shares among green firms

This appendix reports, conditional on a firm having any green innovation, the fraction that also has a labor-augmenting introduction, the fraction with a labor-saving introduction, and the fraction with neither. Product introductions are on the left and process introductions on the right; within each category the faded bar is the regression sample (309k firms) and the solid bar is the full snapshot (605k firms). The denominator is firms with at least one green product (left) or process (right) introduction, so the three categories are not mutually exclusive: a firm can have both an augmenting and a labor-saving introduction, in which case it contributes to both bars.

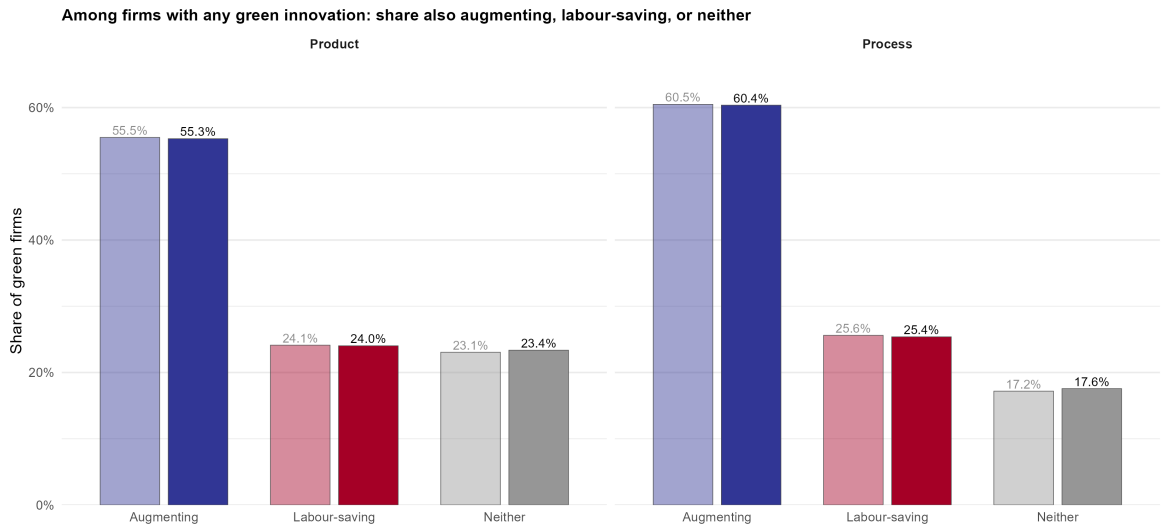


Figure A2: Among firms with any green product (left) or process (right) introduction, the share that also has an augmenting introduction, a labor-saving introduction, or neither. Faded bars: regression sample (309k firms); solid bars: full 605k snapshot. The denominator within each panel is firms with the corresponding green tag. The augmenting and labor-saving categories overlap; "Neither" is firms with the green tag but neither a labor-augmenting nor a labor-saving introduction (typically because the labor effect was classified as "unclear").

E NACE2 sector-level results

This appendix repeats the firm-level direction-tag analysis with each dot representing one NACE2 sector rather than one country. The sector set is the same in every figure: NACE2 codes with at least 100 firms in the regression sample. Dot labels are short verbal descriptors of the NACE Rev. 2 division (e.g. "Wholesale trade", "Software & IT", "Pharmaceuticals"). The aggregations otherwise follow the country-level construction in Sections 4–4.2.

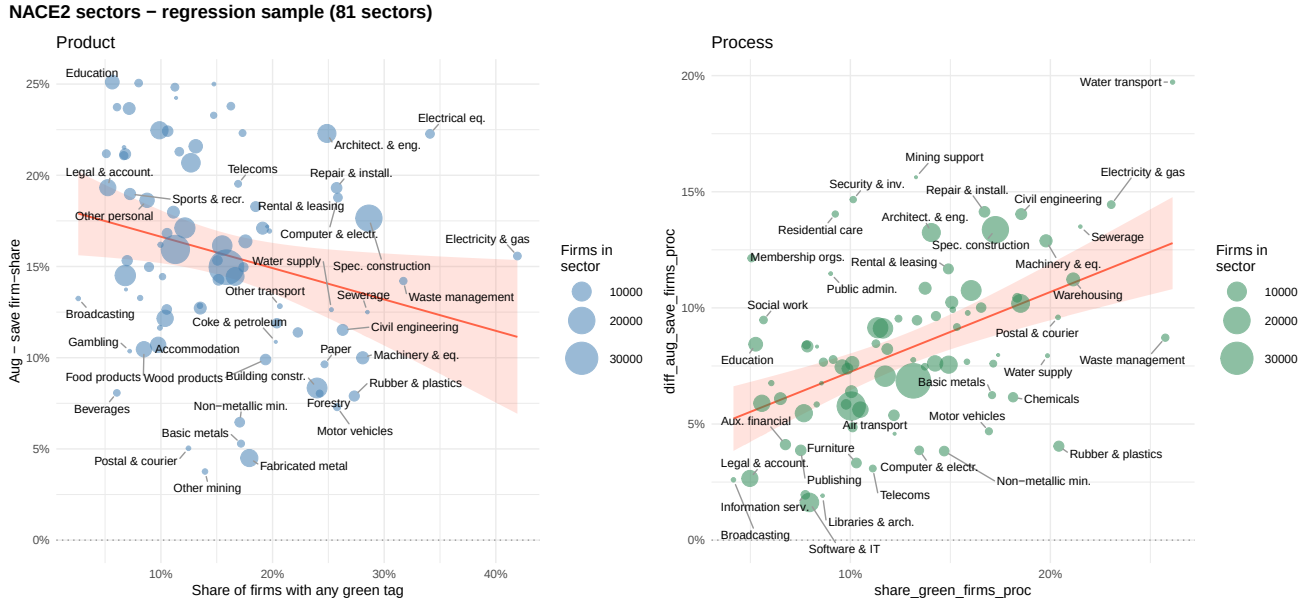


Figure A3: Sector-level direction of innovation, regression sample. Each dot is one NACE2 sector with at least 100 firms in the regression sample. x: share of firms in the sector with any green product (left) or process (right) introduction. y: augmenting firm-share minus labor-saving firm-share. Red line is OLS fit with 95% confidence band.



Figure A4: Sector-level direction of innovation, full 605k snapshot. Same sector set and axes as Figure A3; firms with no extracted text contribute zeros to the firm-tag shares, so the x-axis levels are mechanically lower.

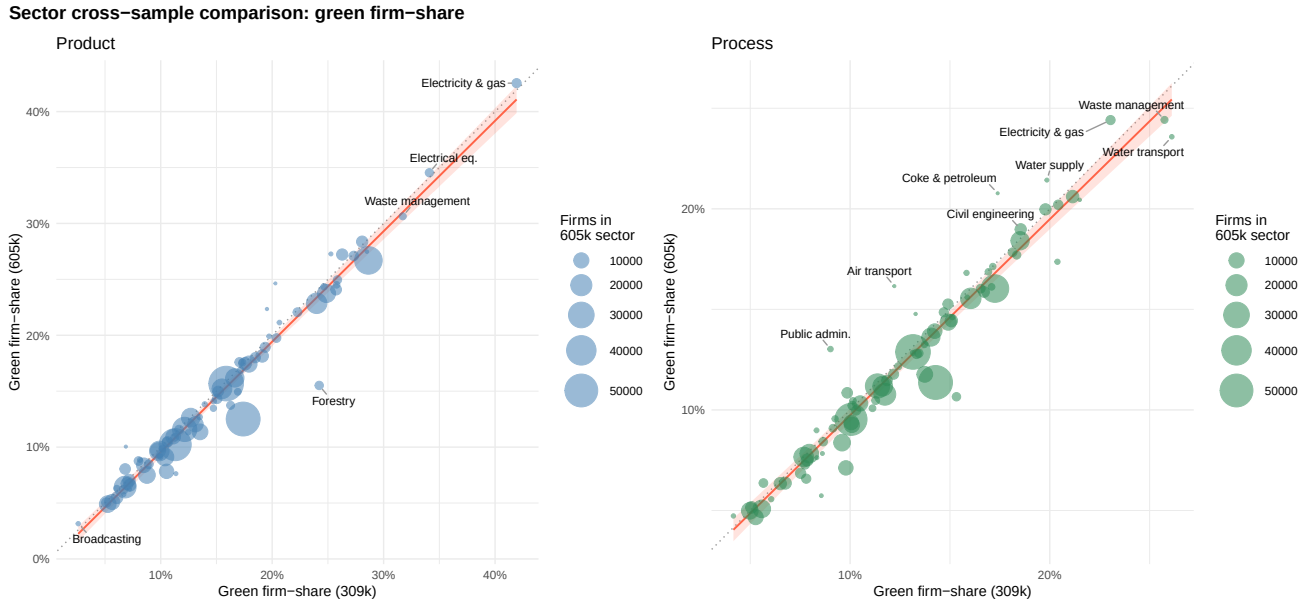


Figure A5: Sector-level cross-sample comparison: green firm-share in the 309k regression sample (x-axis) vs the 605k full snapshot (y-axis). Product (left) and process (right). Dotted line is the 45-degree equality reference.

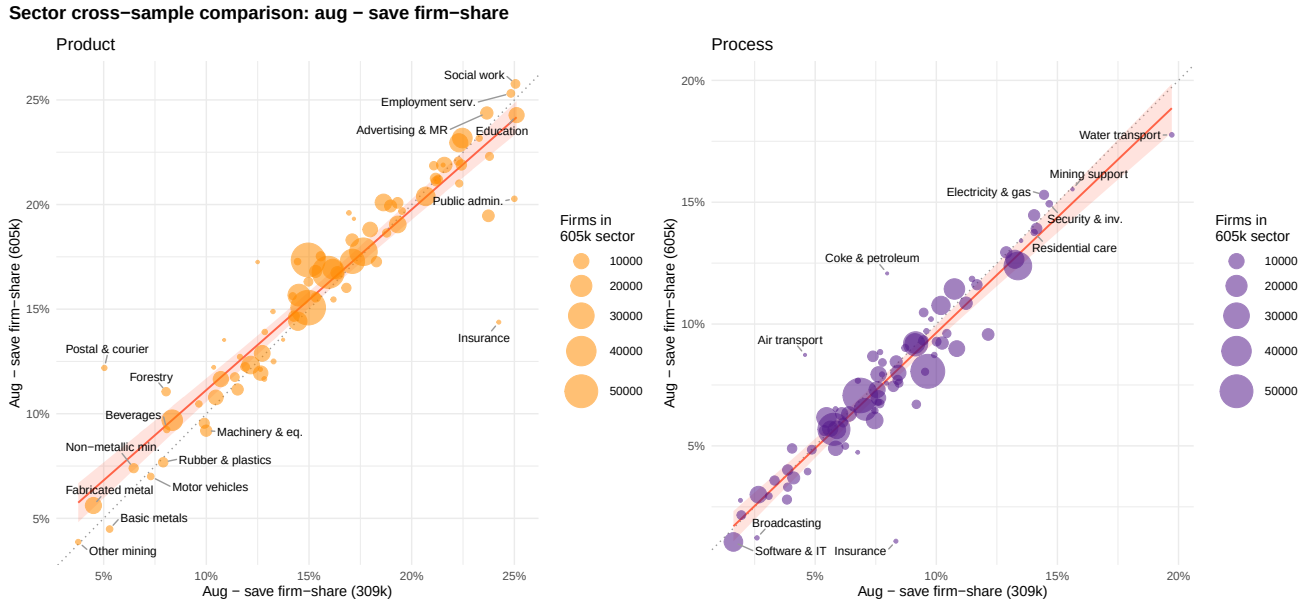


Figure A6: Sector-level cross-sample comparison: aug − save firm-share in the 309k regression sample (x-axis) vs the 605k full snapshot (y-axis). Product (left) and process (right).

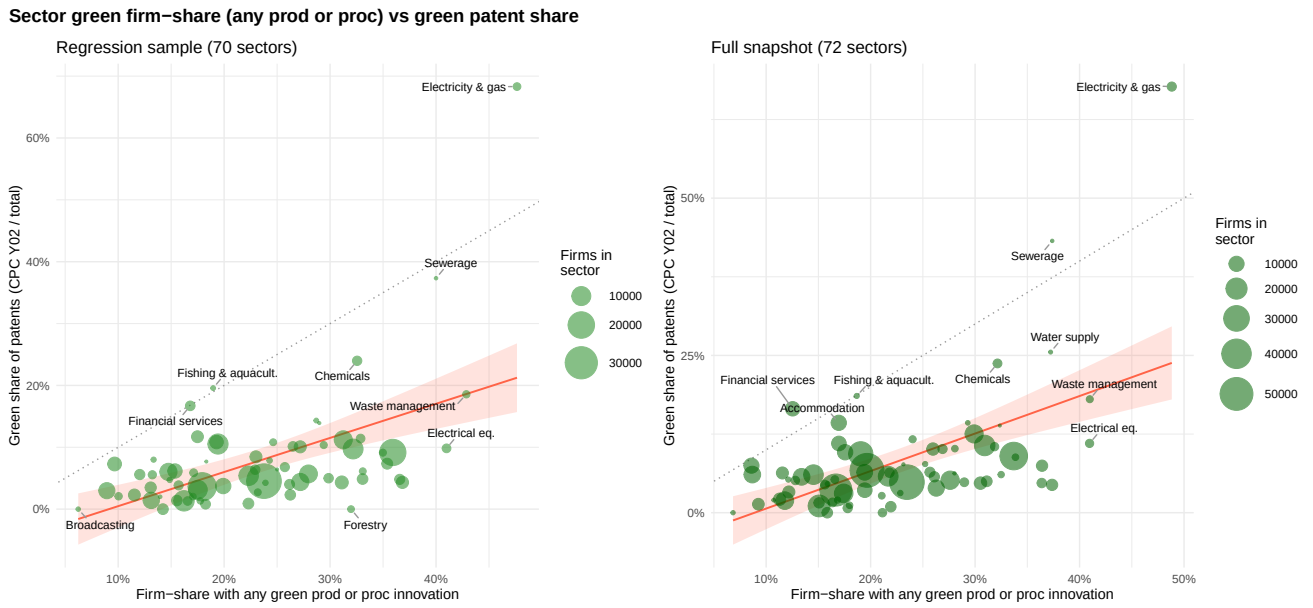


Figure A7: Sector-level green firm-share (any product or process) vs green patent share. Regression sample (left) and full 605k snapshot (right). Restricted to sectors with at least 25 patents and at least 100 firms in the regression sample.

F Country × NACE2 cell-level results

This appendix repeats the firm-level direction-tag analysis at the (country × NACE2) cell level. Each dot is now one country-sector cell rather than one country. With ∼2,700 such cells in the regression sample, individual labels would be unreadable, so we drop text labels and rely on dot density. The cell set is the same in every figure: cells with at least 25 firms in the regression sample.

The cell-level patterns echo the country-level ones in Sections 4–4.2: the cross-cell relationship between greenness and the augmenting–saving net-labor direction is positive and statistically informative; cell-level augmenting and labor-saving shares co-move tightly; and the LLM-extracted greenness–patent greenness relationship is positive and below the 45-degree line in most cells. The within-country sectoral variation captured here is informative about whether country-level patterns are driven by sectoral composition or by within-sector firm heterogeneity.



Figure A8: Cell-level direction of innovation, regression sample. Each dot is one (country, NACE2) cell with at least 25 firms in the regression sample. x: share of firms in the cell with any green product (left) or process (right) introduction. y: augmenting firm-share minus labor-saving firm-share. Red line is OLS fit with 95% confidence band.

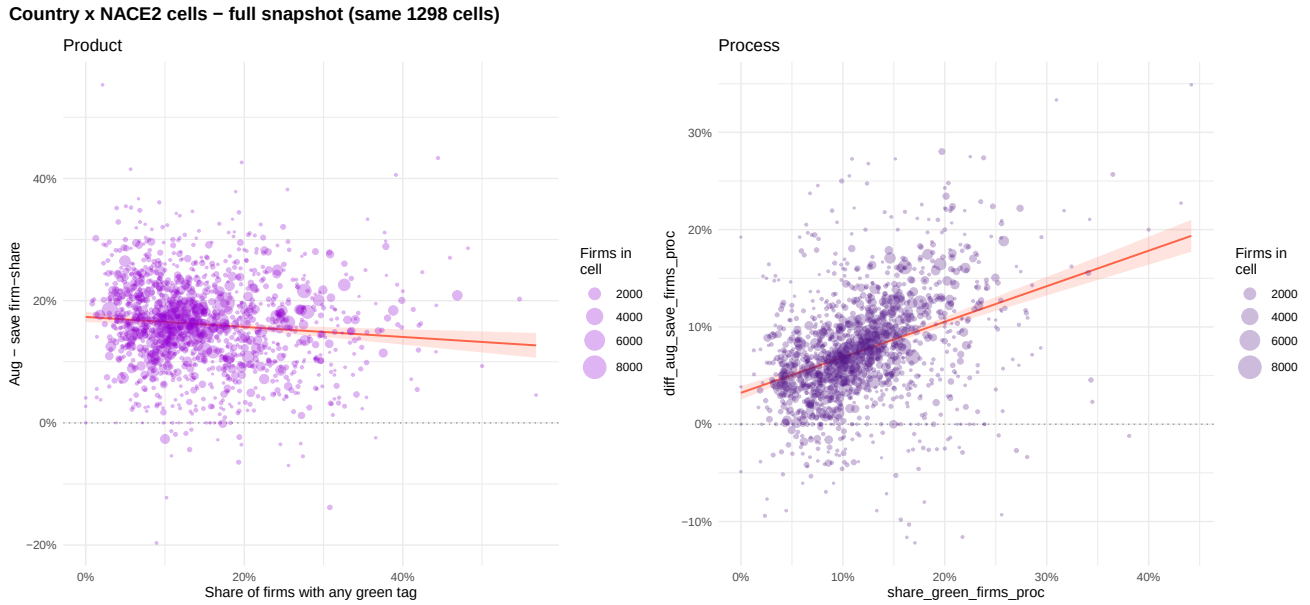


Figure A9: Cell-level direction of innovation, full 605k snapshot. Same cell set and axes as Figure A8; firms with no extracted text contribute zeros to the firm-tag shares, so the x-axis levels are mechanically lower.

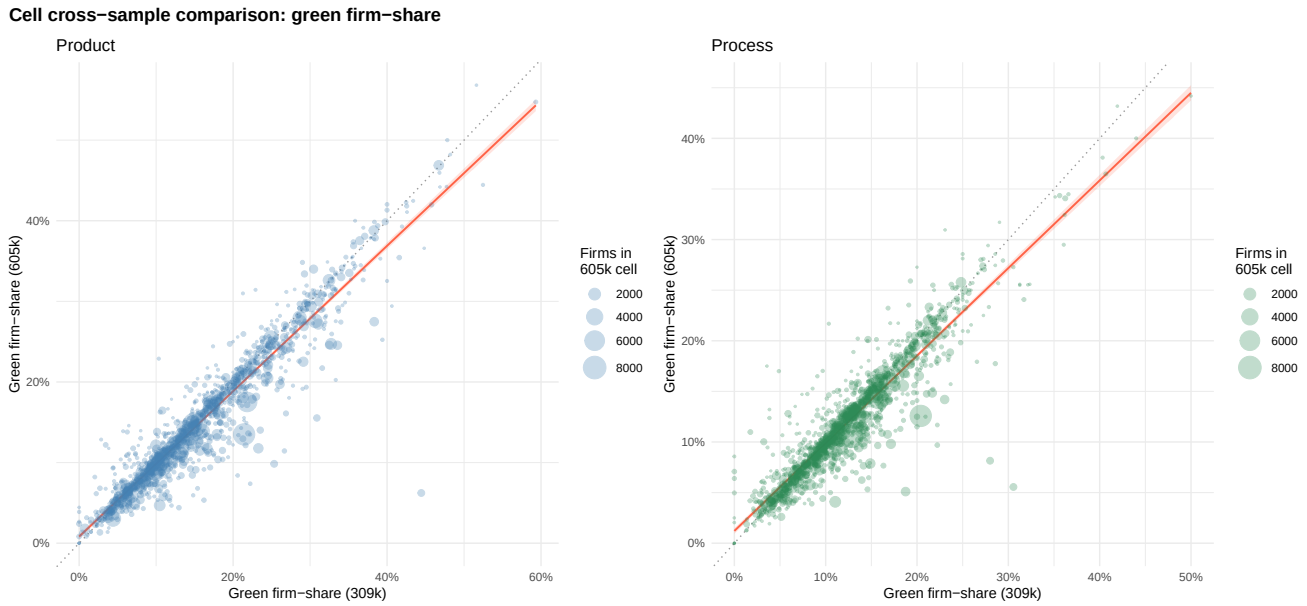


Figure A10: Cell-level cross-sample comparison: green firm-share in the 309k regression sample (x-axis) vs the 605k full snapshot (y-axis). Product (left) and process (right). Dotted line is the 45-degree equality reference.

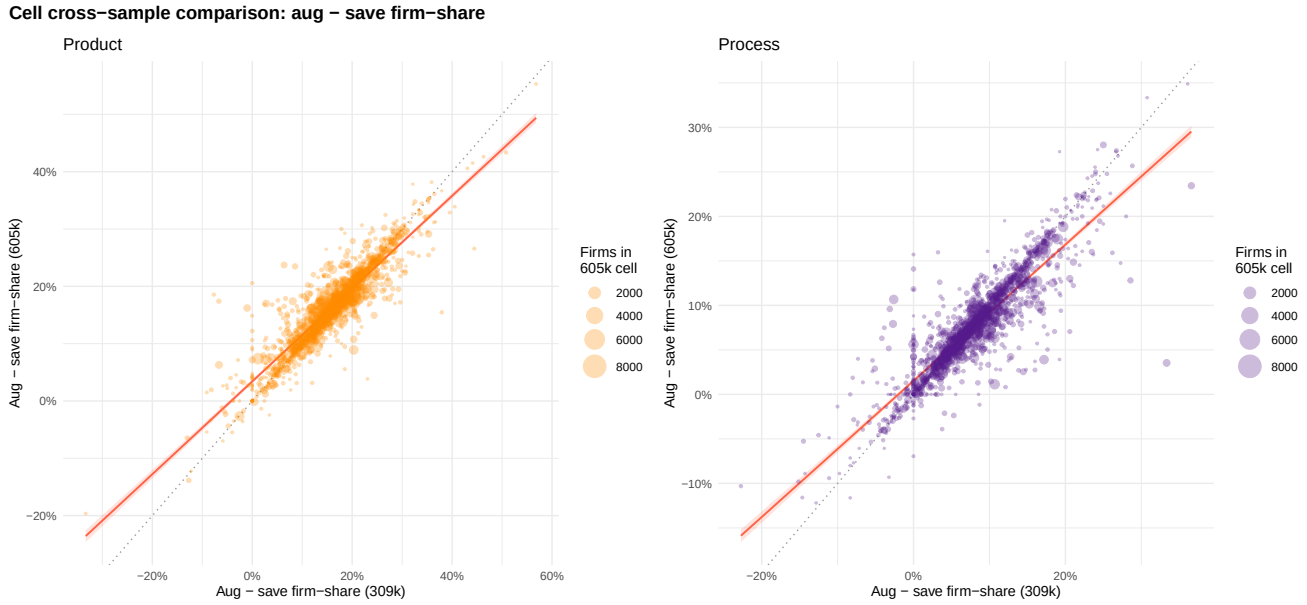


Figure A11: Cell-level cross-sample comparison: aug − save firm-share in the 309k regression sample (x-axis) vs the 605k full snapshot (y-axis). Product (left) and process (right).



Figure A12: Cell-level green firm-share (any product or process) vs green patent share. Regression sample (left) and full 605k snapshot (right). Restricted to cells with at least 5 patents and at least 25 firms in the regression sample.

G Sector vs country variance decomposition

This appendix decomposes the cross-firm variation in the six direction tags into a country part and a NACE2 sector part. Table A2 reports R^2 shares; Figure A13 compares raw country shares to the corresponding sector-adjusted shares.

Table A2: Variance decomposition of firm-level direction tags.

Tag	Country	Sector	Country + Sector	Interaction extra	Residual
Green (product)	0.8	3.5	4.3	1.1	94.7
Green (process)	0.5	1.4	1.8	0.8	97.3
Augmenting (product)	0.4	0.8	1.2	0.8	98.0
Augmenting (process)	0.5	0.6	1.2	0.8	98.0
Labor-saving (product)	0.3	1.1	1.4	0.8	97.8
Labor-saving (process)	0.3	0.4	0.7	0.7	98.6

Notes: Rows are firm-level binary direction tags; entries are R^2 contributions in percent from a linear-probability regression of the tag on country and NACE2 sector fixed effects. The "Country" and "Sector" columns report the Shapley share of each fixed-effect block (the order-free average of the two sequential orderings). "Country + Sector" is the joint R^2 from the additive two-way fixed-effects specification. "Interaction extra" is the additional R^2 contributed by the full country×sector interaction beyond the additive model. "Residual" is the remaining within-cell variation. Sample: 553,048 firms across 40 countries and 86 NACE2 sectors, restricted to countries with at least 25 firms in the regression sample.

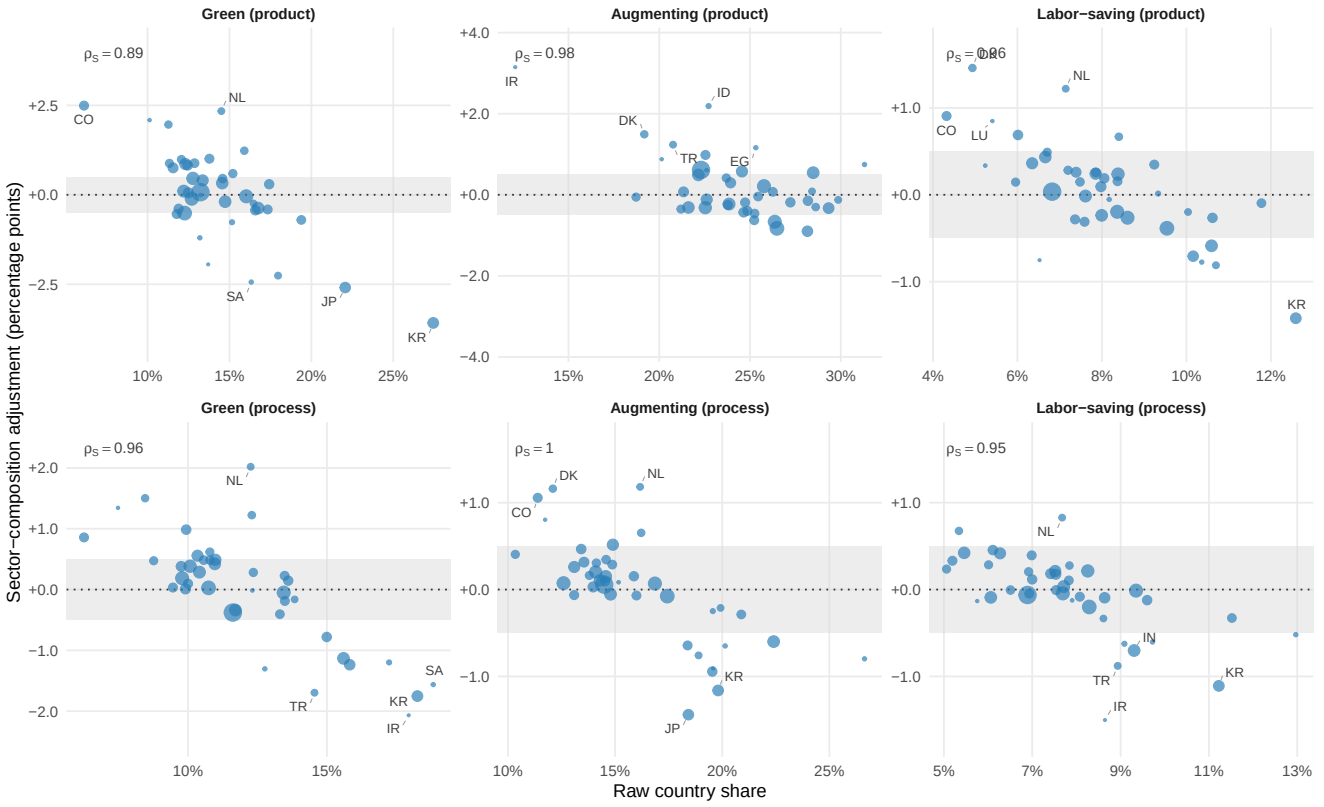


Figure A13: Sector-composition adjustment to country direction-tag shares. Each dot is a country; the y-axis is the sector-adjusted country share minus the raw country share, in percentage points. Positive values mean adjusting for sector mix raises the country's share; negative values mean the raw share was inflated by sector composition. The dotted reference line at zero marks no adjustment, and the grey band marks ± 0.5 pp. The Spearman rank correlation between raw and adjusted shares appears in the top-left of each panel. Country labels mark the five countries with the largest absolute adjustment in each panel.

Notes: The adjusted country share is constructed by taking the firm-level residual from a regression of the binary direction tag on NACE2 sector fixed effects, averaging within country, and adding back the grand mean. The construction isolates the within-sector cross-country contribution to the country share. Sample as in Table A2. Dot size is proportional to firm count in the country.